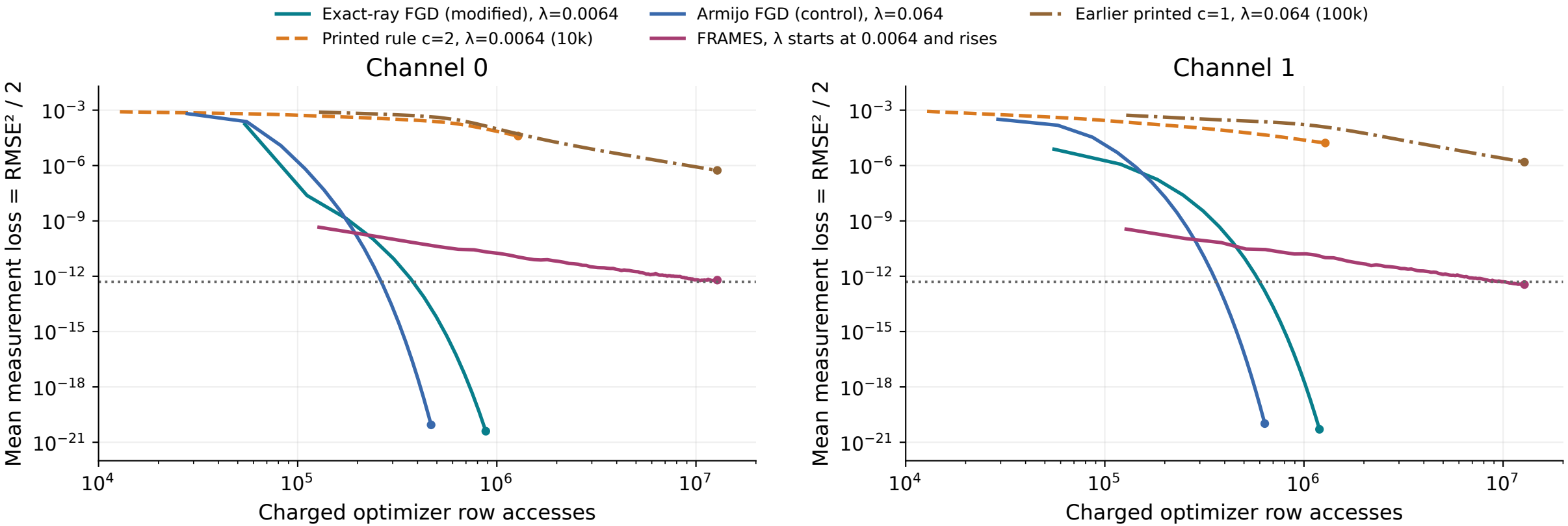
Current Quiroga-style setup: measurement loss

Two qubits · noiseless · 128 rows · shared target, data and initialization within each channel



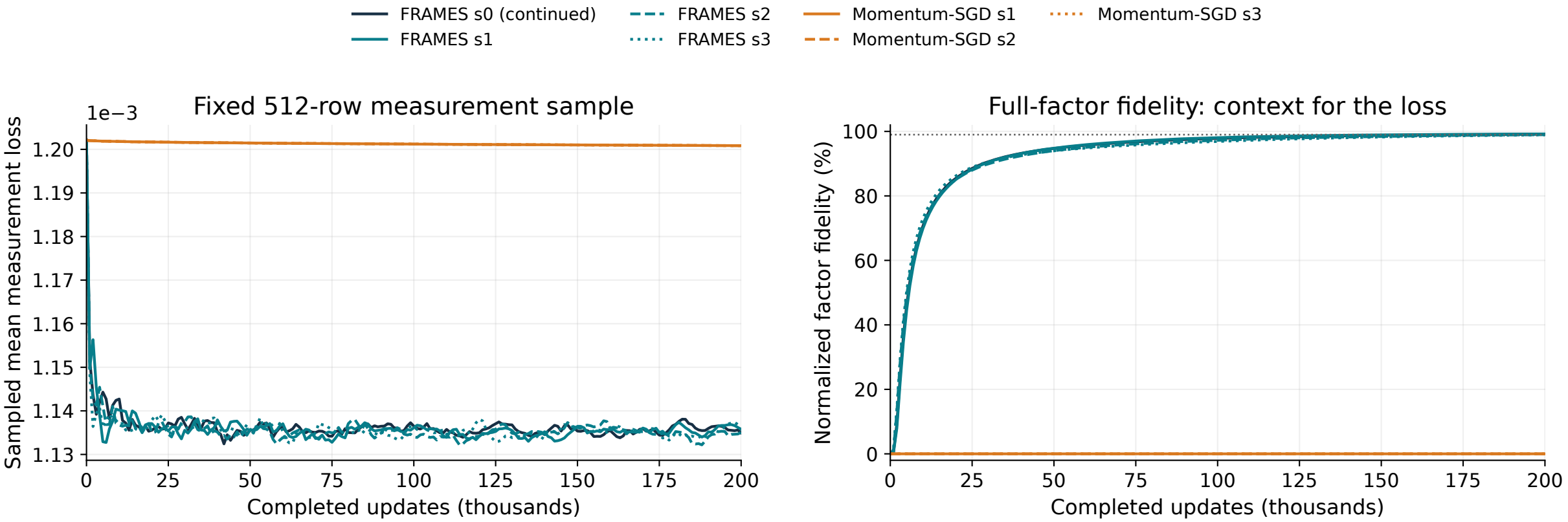
Dotted line: measurement-loss threshold 5×10^{-13}, equivalent to RMSE $\le 10^{-6}$. Passing this line alone does not establish the joint fidelity/TP gate. Initial zero-work checkpoints are omitted on the logarithmic x axis.

Loss excludes TP penalties. Penalties and update horizons differ; these are paired diagnostics, not a final best-for-best comparison. Exact-ray and Armijo are modified/control methods, not the printed adaFGD rule.

Charged rows include line-search evaluations but exclude spectral-norm, TP and checkpoint work. This is not a wall-time comparison.

Earlier local-Pauli setup: eight-qubit loss

Gaussian observation noise σ=0.05 · one channel / initialization · sampling streams s0–s3



All loss curves use the same fixed 512-row diagnostic sample, not the full virtual dataset. The y axis is zoomed and does not start at zero. Noise-dominated loss values can be close while recovered channels differ greatly.

FRAMES ends at 98.92–99.22% fidelity with normalized TP ≈0.0107. Momentum-SGD stays near 0.000873% fidelity in these tested settings. These are sampling replicates, not independent target channels.

The s0 curve joins saved history at 148,000 updates to its verified continuation. Same batch size 65,536 throughout; no new runs. This noisy local-Pauli benchmark is separate from the first page.